

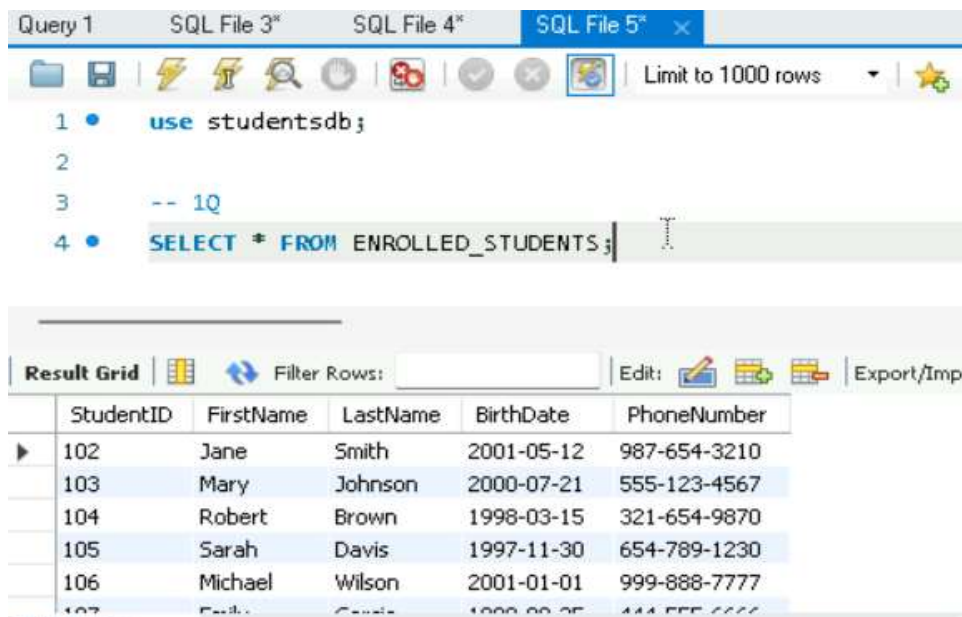
Assignment - 19

1. Write a query to retrieve all columns and records from the ENROLLED_STUDENTS table.

use studentsdb;

-- 1Q

SELECT * FROM ENROLLED_STUDENTS;



The screenshot shows a SQL Server Enterprise Manager interface. At the top, there are tabs for 'Query 1', 'SQL File 3*', 'SQL File 4*', and 'SQL File 5*'. Below the tabs is a toolbar with various icons. The main area displays a query window with the following text:

```
1 • use studentsdb;
2
3 -- 1Q
4 • SELECT * FROM ENROLLED_STUDENTS;
```

Below the query window is a 'Result Grid' tab. The grid shows the results of the query, with columns: StudentID, FirstName, LastName, BirthDate, and PhoneNumber. The data is as follows:

StudentID	FirstName	LastName	BirthDate	PhoneNumber
102	Jane	Smith	2001-05-12	987-654-3210
103	Mary	Johnson	2000-07-21	555-123-4567
104	Robert	Brown	1998-03-15	321-654-9870
105	Sarah	Davis	1997-11-30	654-789-1230
106	Michael	Wilson	2001-01-01	999-888-7777

2. Write a query to retrieve only the FirstName and LastName columns from the ENROLLED_STUDENTS table.

-- 2Q

SELECT FirstName, LastName FROM ENROLLED_STUDENTS;



The screenshot shows a SQL Server Enterprise Manager interface. At the top, there are tabs for 'Query 1', 'SQL File 3*', 'SQL File 4*', and 'SQL File 5*'. Below the tabs is a toolbar with various icons. The main area displays a query window with the following text:

```
6 -- 2Q
7 • SELECT FirstName, LastName FROM ENROLLED_STUDENTS;
```

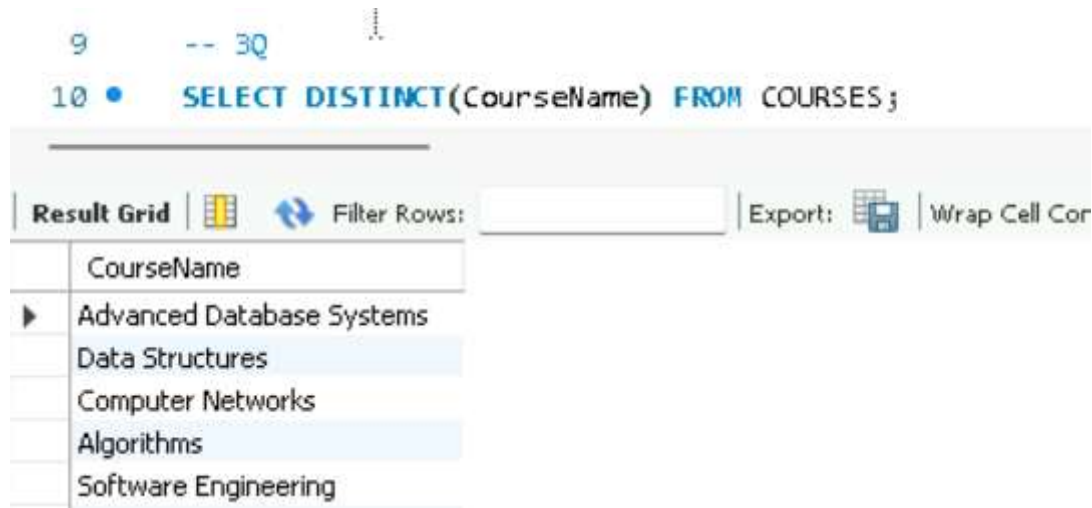
Below the query window is a 'Result Grid' tab. The grid shows the results of the query, with columns: FirstName and LastName. The data is as follows:

FirstName	LastName
Jane	Smith
Mary	Johnson
Robert	Brown
Sarah	Davis
Michael	Wilson

3. Write a query to retrieve all unique CourseName values from the COURSES table.

-- 3Q

```
SELECT DISTINCT(CourseName) FROM COURSES;
```



The screenshot shows a query editor with the following SQL query:

```
9      -- 3Q
10 •   SELECT DISTINCT(CourseName) FROM COURSES;
```

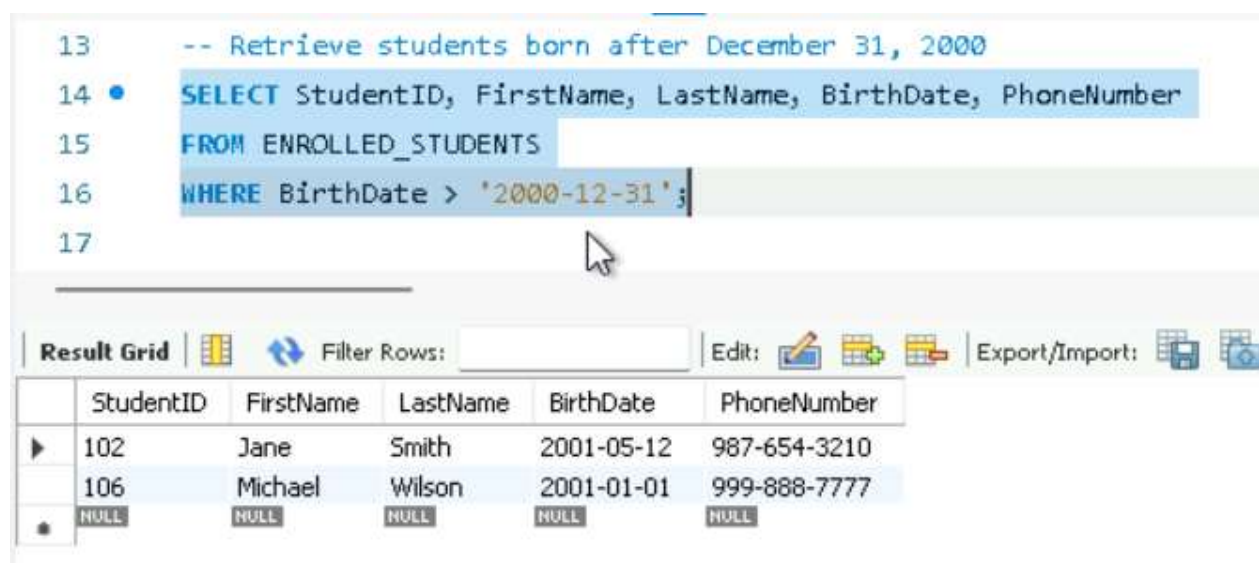
Below the query editor, the 'Result Grid' is displayed with the following data:

CourseName
Advanced Database Systems
Data Structures
Computer Networks
Algorithms
Software Engineering

4. Write a query to retrieve the details of students from the ENROLLED_STUDENTS table who were born after the year 2000.

-- 4Q

```
SELECT StudentID, FirstName, LastName, BirthDate, PhoneNumber
FROM ENROLLED_STUDENTS
WHERE BirthDate > '2000-12-31';
```



The screenshot shows a query editor with the following SQL query:

```
13      -- Retrieve students born after December 31, 2000
14 •   SELECT StudentID, FirstName, LastName, BirthDate, PhoneNumber
15     FROM ENROLLED_STUDENTS
16     WHERE BirthDate > '2000-12-31';
17
```

Below the query editor, the 'Result Grid' is displayed with the following data:

StudentID	FirstName	LastName	BirthDate	PhoneNumber
102	Jane	Smith	2001-05-12	987-654-3210
106	Michael	Wilson	2001-01-01	999-888-7777
NULL	NULL	NULL	NULL	NULL

5. Write a query to retrieve all columns from the COURSES table where the Credits are equal to 3.

-- 5Q

```
SELECT * FROM COURSES
WHERE Credits = 3;
```

```
13  -- Retrieve students born after December 31, 2000
14  • SELECT StudentID, FirstName, LastName, BirthDate, PhoneNumber
15  FROM ENROLLED_STUDENTS
16  WHERE BirthDate > '2000-12-31';
17
```

Result Grid | Filter Rows: | Edit: | Export/Import:

	StudentID	FirstName	LastName	BirthDate	PhoneNumber
▶	102	Jane	Smith	2001-05-12	987-654-3210
	106	Michael	Wilson	2001-01-01	999-888-7777
•	NULL	NULL	NULL	NULL	NULL

6. Write a query to retrieve the details of students from ENROLLED_STUDENTS where the PhoneNumber is '123-456-7890'.

-- 6Q

```
SELECT StudentID, FirstName, LastName, BirthDate, PhoneNumber
FROM ENROLLED_STUDENTS
WHERE PhoneNumber = '123-456-7890';
```

```
23  • SELECT StudentID, FirstName, LastName, BirthDate, PhoneNumber
24  FROM ENROLLED_STUDENTS
25  WHERE PhoneNumber = '123-456-7890';
26
```

Result Grid | Filter Rows: | Edit: | Export/Import:

	StudentID	FirstName	LastName	BirthDate	PhoneNumber
•	NULL	NULL	NULL	NULL	NULL

7. Write a query to retrieve the details of students from ENROLLED_STUDENTS where the LastName starts with the letter 'S'.

-- 7Q

```
SELECT StudentID, FirstName, LastName, BirthDate, PhoneNumber
FROM ENROLLED_STUDENTS
WHERE LastName LIKE 'S%';
```

```
28  -- Retrieve students whose last name begins with the letter S
29  •  SELECT StudentID, FirstName, LastName, BirthDate, PhoneNumber
30  FROM ENROLLED_STUDENTS
31  WHERE LastName LIKE 'S%';
32
```

Result Grid

	StudentID	FirstName	LastName	BirthDate	PhoneNumber
▶	102	Jane	Smith	2001-05-12	987-654-3210
*	NULL	NULL	NULL	NULL	NULL

8. Write a query to retrieve all columns from the COURSES table, ordered by the Credits in ascending order.

-- 8Q

```
SELECT * FROM COURSES
ORDER BY Credits ASC;
```

```
33  -- 8Q
34  •  SELECT * FROM COURSES
35  ORDER BY Credits ASC;
36
37
```

Limit to 1000

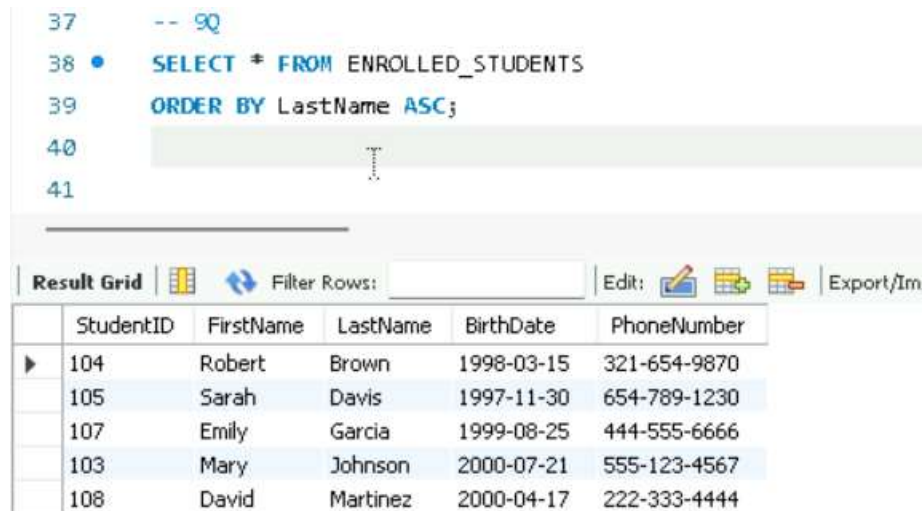
Result Grid

	CourseID	CourseName	Credits
▶	206	Software Engineering	2
	201	Advanced Database Systems	3
	203	Data Structures	3
	204	Computer Networks	3
	207	Web Development	3

9. Write a query to retrieve all columns from the ENROLLED_STUDENTS table, ordered alphabetically by LastName.

-- 9Q

```
SELECT * FROM ENROLLED_STUDENTS  
ORDER BY LastName ASC;
```



The screenshot shows a SQL query editor with the following code:

```
37 -- 9Q  
38 • SELECT * FROM ENROLLED_STUDENTS  
39 ORDER BY LastName ASC;  
40  
41
```

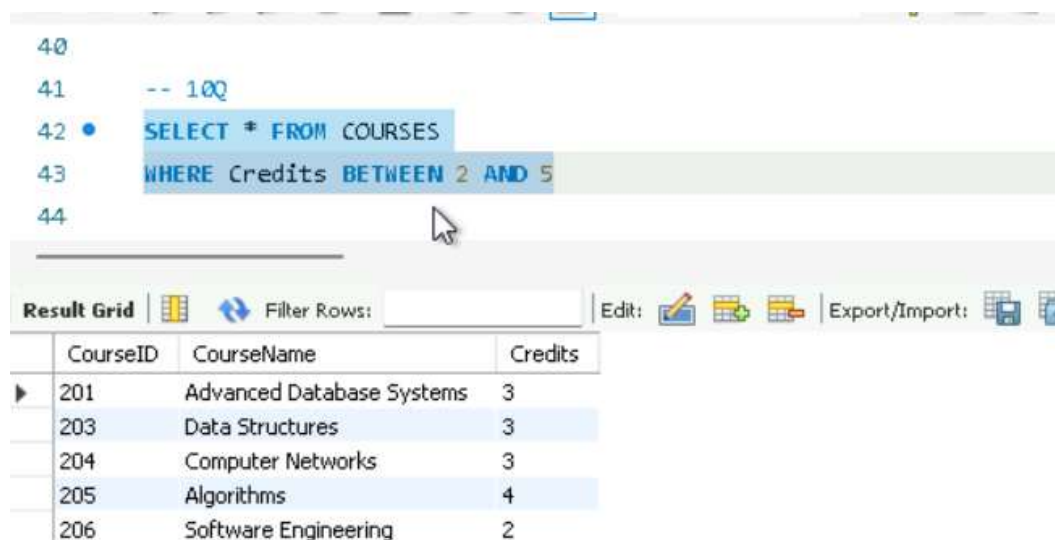
Below the editor is a toolbar with options: Result Grid, Filter Rows, Edit, and Export/Import. The result grid displays the following data:

	StudentID	FirstName	LastName	BirthDate	PhoneNumber
▶	104	Robert	Brown	1998-03-15	321-654-9870
	105	Sarah	Davis	1997-11-30	654-789-1230
	107	Emily	Garcia	1999-08-25	444-555-6666
	103	Mary	Johnson	2000-07-21	555-123-4567
	108	David	Martinez	2000-04-17	222-333-4444

10. Write a query to retrieve the details of courses from the COURSES table where the Credits are between 2 and 5 (inclusive).

-- 10Q

```
SELECT * FROM COURSES  
WHERE Credits BETWEEN 2 AND 5;
```



The screenshot shows a SQL query editor with the following code:

```
40  
41 -- 10Q  
42 • SELECT * FROM COURSES  
43 WHERE Credits BETWEEN 2 AND 5  
44
```

Below the editor is a toolbar with options: Result Grid, Filter Rows, Edit, and Export/Import. The result grid displays the following data:

	CourseID	CourseName	Credits
▶	201	Advanced Database Systems	3
	203	Data Structures	3
	204	Computer Networks	3
	205	Algorithms	4
	206	Software Engineering	2

11. Write a query to retrieve the details of students from ENROLLED_STUDENTS where the PhoneNumber is NULL.

-- 11Q

```
SELECT * FROM ENROLLED_STUDENTS  
WHERE PhoneNumber = NULL;
```

The screenshot shows a SQL query editor with the following text:

```
45 -- 11Q  
46 • SELECT * FROM ENROLLED_STUDENTS  
47 WHERE PhoneNumber = NULL;  
48  
49
```

Below the editor is a toolbar with icons for 'Result Grid', 'Filter Rows', 'Edit', and 'Export/Import'. The 'Result Grid' is active, displaying a table with the following data:

StudentID	FirstName	LastName	BirthDate	PhoneNumber
NULL	NULL	NULL	NULL	NULL

12. Write a query to retrieve the first three students with the smallest StudentID from the ENROLLED_STUDENTS table.

-- 12Q

```
SELECT * FROM ENROLLED_STUDENTS  
ORDER BY StudentID ASC  
LIMIT 3;
```

The screenshot shows a SQL query editor with the following text:

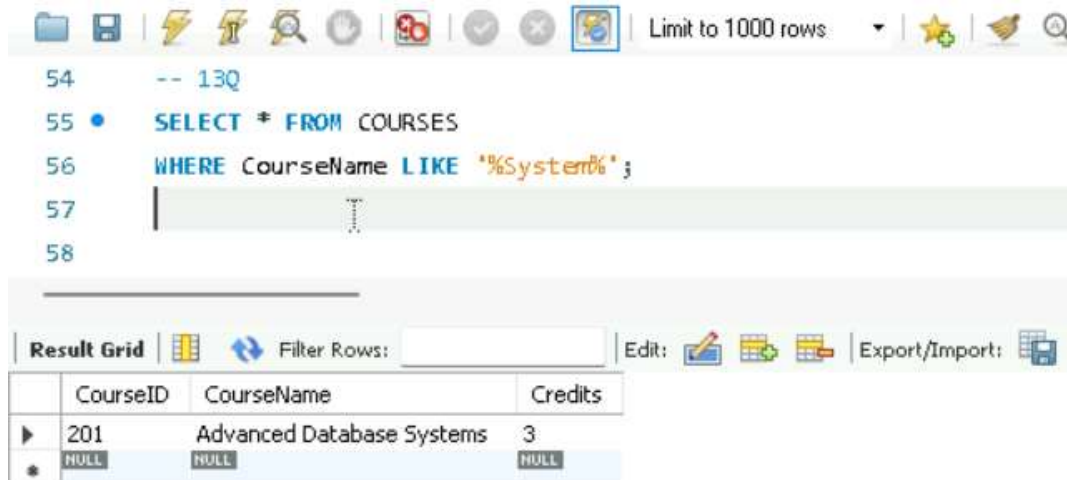
```
48  
49 -- 12Q  
50 • SELECT * FROM ENROLLED_STUDENTS  
51 ORDER BY StudentID ASC  
52 LIMIT 3;
```

Below the editor is a toolbar with icons for 'Result Grid', 'Filter Rows', 'Edit', and 'Export/Import'. The 'Result Grid' is active, displaying a table with the following data:

StudentID	FirstName	LastName	BirthDate	PhoneNumber
102	Jane	Smith	2001-05-12	987-654-3210
103	Mary	Johnson	2000-07-21	555-123-4567
104	Robert	Brown	1998-03-15	321-654-9870
NULL	NULL	NULL	NULL	NULL

13. Write a query to retrieve all columns from the COURSES table where the CourseName contains the word 'System'.

```
-- 13Q
SELECT * FROM COURSES
WHERE CourseName LIKE '%System%';
```



The screenshot shows a database query editor with a toolbar at the top. The SQL query is entered in the editor area:

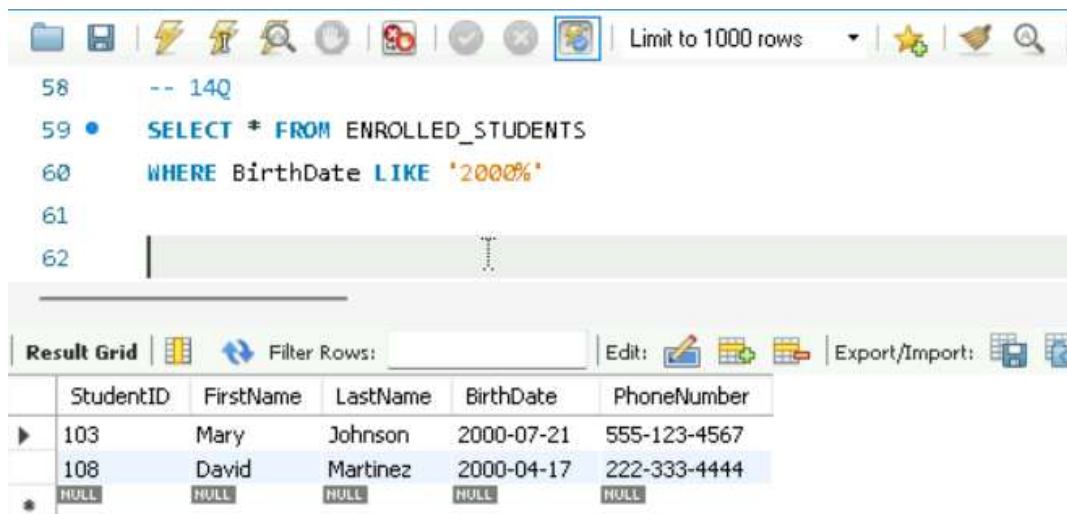
```
-- 13Q
SELECT * FROM COURSES
WHERE CourseName LIKE '%System%';
```

Below the query editor, the 'Result Grid' is displayed. It has a toolbar with 'Filter Rows', 'Edit', and 'Export/Import' options. The grid contains the following data:

CourseID	CourseName	Credits
201	Advanced Database Systems	3
NULL	NULL	NULL

14. Write a query to retrieve the details of students from the ENROLLED_STUDENTS table who were born in the year 2000.

```
-- 14Q
SELECT * FROM ENROLLED_STUDENTS
WHERE BirthDate LIKE '2000%';
```



The screenshot shows a database query editor with a toolbar at the top. The SQL query is entered in the editor area:

```
-- 14Q
SELECT * FROM ENROLLED_STUDENTS
WHERE BirthDate LIKE '2000%';
```

Below the query editor, the 'Result Grid' is displayed. It has a toolbar with 'Filter Rows', 'Edit', and 'Export/Import' options. The grid contains the following data:

StudentID	FirstName	LastName	BirthDate	PhoneNumber
103	Mary	Johnson	2000-07-21	555-123-4567
108	David	Martinez	2000-04-17	222-333-4444
NULL	NULL	NULL	NULL	NULL

15. Write a query to retrieve the details of students from the ENROLLED_STUDENTS table where the FirstName is 'John' and the LastName is 'Doe'.

-- 15Q

```
SELECT StudentID, FirstName, LastName, BirthDate, PhoneNumber
FROM ENROLLED_STUDENTS
WHERE FirstName = 'John' AND LastName = 'Doe';
```

The screenshot shows a database query editor with a toolbar at the top. The SQL query is entered in the main text area:

```
62 -- 15Q
63 SELECT StudentID, FirstName, LastName, BirthDate, PhoneNumber
64 FROM ENROLLED_STUDENTS
65 WHERE FirstName = 'John' AND LastName = 'Doe';
66
```

Below the query editor is the 'Result Grid' section. It has a toolbar with 'Filter Rows', 'Edit', and 'Export/Import' options. The result grid shows a single row with all columns set to NULL.

	StudentID	FirstName	LastName	BirthDate	PhoneNumber
*	NULL	NULL	NULL	NULL	NULL

16. Write a query to retrieve all columns from the COURSES table, ordered by CourseName alphabetically.

-- 16Q

```
SELECT * FROM COURSES
ORDER BY CourseName ASC;
```

The screenshot shows a database query editor with a toolbar at the top. The SQL query is entered in the main text area:

```
67 -- 16Q
68 SELECT *
69 FROM COURSES
70 ORDER BY CourseName ASC;
71
```

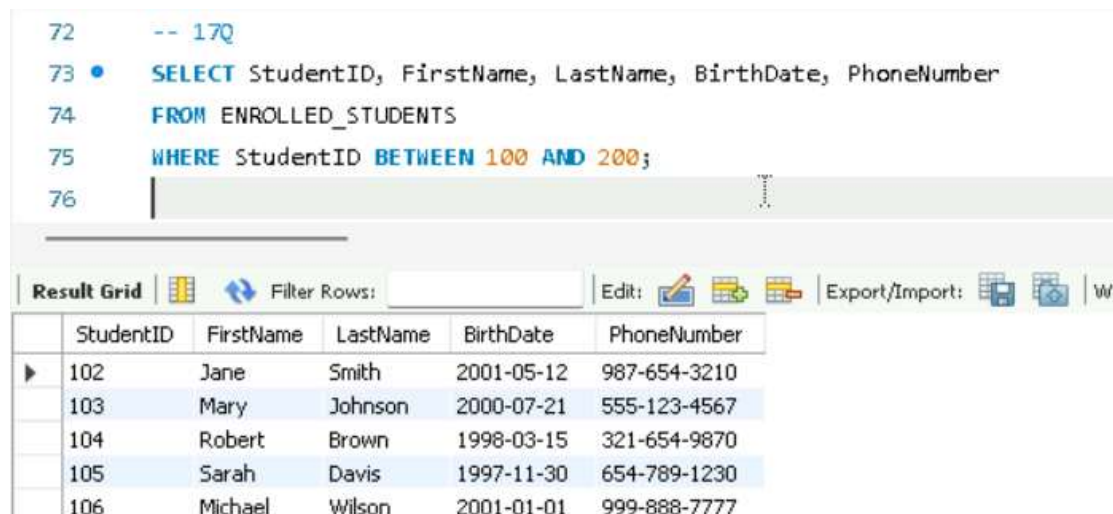
Below the query editor is the 'Result Grid' section. It has a toolbar with 'Filter Rows', 'Edit', and 'Export/Import' options. The result grid shows a list of courses ordered alphabetically by CourseName.

	CourseID	CourseName	Credits
▶	201	Advanced Database Systems	3
	205	Algorithms	4
	204	Computer Networks	3
	203	Data Structures	3
	208	Machine Learning	5

17. Write a query to retrieve the details of students from the ENROLLED_STUDENTS table where the StudentID is between 100 and 200.

-- 17Q

```
SELECT StudentID, FirstName, LastName, BirthDate, PhoneNumber
FROM ENROLLED_STUDENTS
WHERE StudentID BETWEEN 100 AND 200;
```



The screenshot shows a SQL query editor with the following query:

```
-- 17Q
SELECT StudentID, FirstName, LastName, BirthDate, PhoneNumber
FROM ENROLLED_STUDENTS
WHERE StudentID BETWEEN 100 AND 200;
```

Below the query editor is a "Result Grid" showing the results of the query. The grid has columns for StudentID, FirstName, LastName, BirthDate, and PhoneNumber. The results are as follows:

StudentID	FirstName	LastName	BirthDate	PhoneNumber
102	Jane	Smith	2001-05-12	987-654-3210
103	Mary	Johnson	2000-07-21	555-123-4567
104	Robert	Brown	1998-03-15	321-654-9870
105	Sarah	Davis	1997-11-30	654-789-1230
106	Michael	Wilson	2001-01-01	999-888-7777

18. Write a query to retrieve the details of students from the ENROLLED_STUDENTS table who were born in the month of December.

-- 18Q

```
SELECT StudentID, FirstName, LastName, BirthDate, PhoneNumber
FROM ENROLLED_STUDENTS
WHERE MONTH(BirthDate) = 12;
```



The screenshot shows a SQL query editor with the following query:

```
-- 18Q
SELECT StudentID, FirstName, LastName, BirthDate, PhoneNumber
FROM ENROLLED_STUDENTS
WHERE MONTH(BirthDate) = 12;
```

Below the query editor is a "Result Grid" showing the results of the query. The grid has columns for StudentID, FirstName, LastName, BirthDate, and PhoneNumber. The results are all NULL, indicating that no students were found born in December.

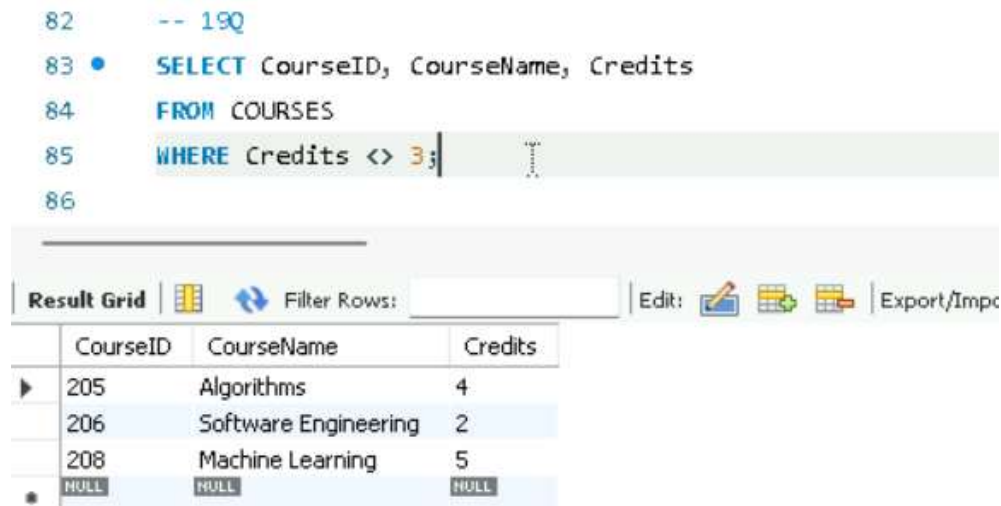
StudentID	FirstName	LastName	BirthDate	PhoneNumber
NULL	NULL	NULL	NULL	NULL

19. Write a query to retrieve all columns from the COURSES table where the Credits are not equal to 3.

-- 19Q

```
SELECT CourseID, CourseName, Credits
FROM COURSES
WHERE Credits <> 3;
```

```
82      -- 19Q
83 •    SELECT CourseID, CourseName, Credits
84      FROM COURSES
85      WHERE Credits <> 3;
86
```



The screenshot shows a database query editor with a SQL query and its results. The query is: `SELECT CourseID, CourseName, Credits FROM COURSES WHERE Credits <> 3;`. The results are displayed in a grid with columns CourseID, CourseName, and Credits. The results are:

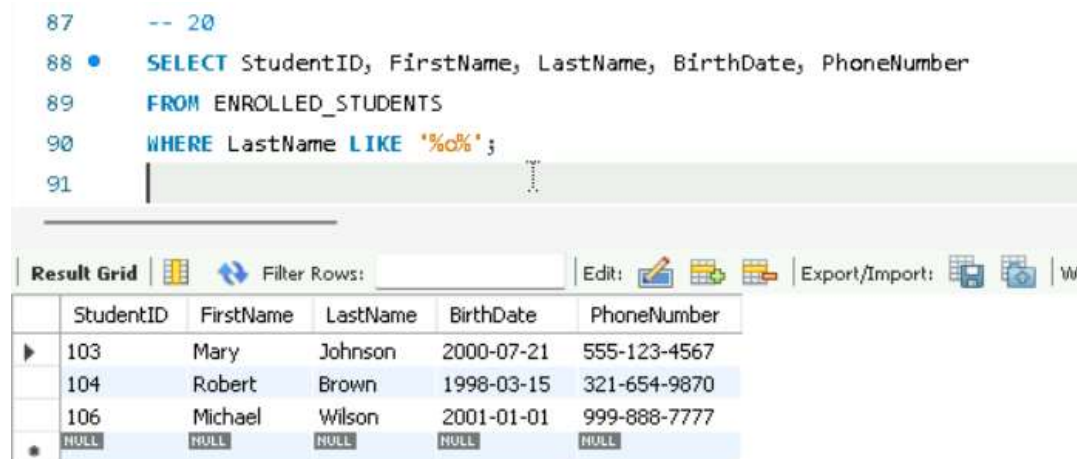
CourseID	CourseName	Credits
205	Algorithms	4
206	Software Engineering	2
208	Machine Learning	5
NULL	NULL	NULL

20. Write a query to retrieve the details of students from the ENROLLED_STUDENTS table where the LastName contains the letter 'o'.

-- 20

```
SELECT StudentID, FirstName, LastName, BirthDate, PhoneNumber
FROM ENROLLED_STUDENTS
WHERE LastName LIKE '%o%';
```

```
87      -- 20
88 •    SELECT StudentID, FirstName, LastName, BirthDate, PhoneNumber
89      FROM ENROLLED_STUDENTS
90      WHERE LastName LIKE '%o%';
91
```



The screenshot shows a database query editor with a SQL query and its results. The query is: `SELECT StudentID, FirstName, LastName, BirthDate, PhoneNumber FROM ENROLLED_STUDENTS WHERE LastName LIKE '%o%';`. The results are displayed in a grid with columns StudentID, FirstName, LastName, BirthDate, and PhoneNumber. The results are:

StudentID	FirstName	LastName	BirthDate	PhoneNumber
103	Mary	Johnson	2000-07-21	555-123-4567
104	Robert	Brown	1998-03-15	321-654-9870
106	Michael	Wilson	2001-01-01	999-888-7777
NULL	NULL	NULL	NULL	NULL